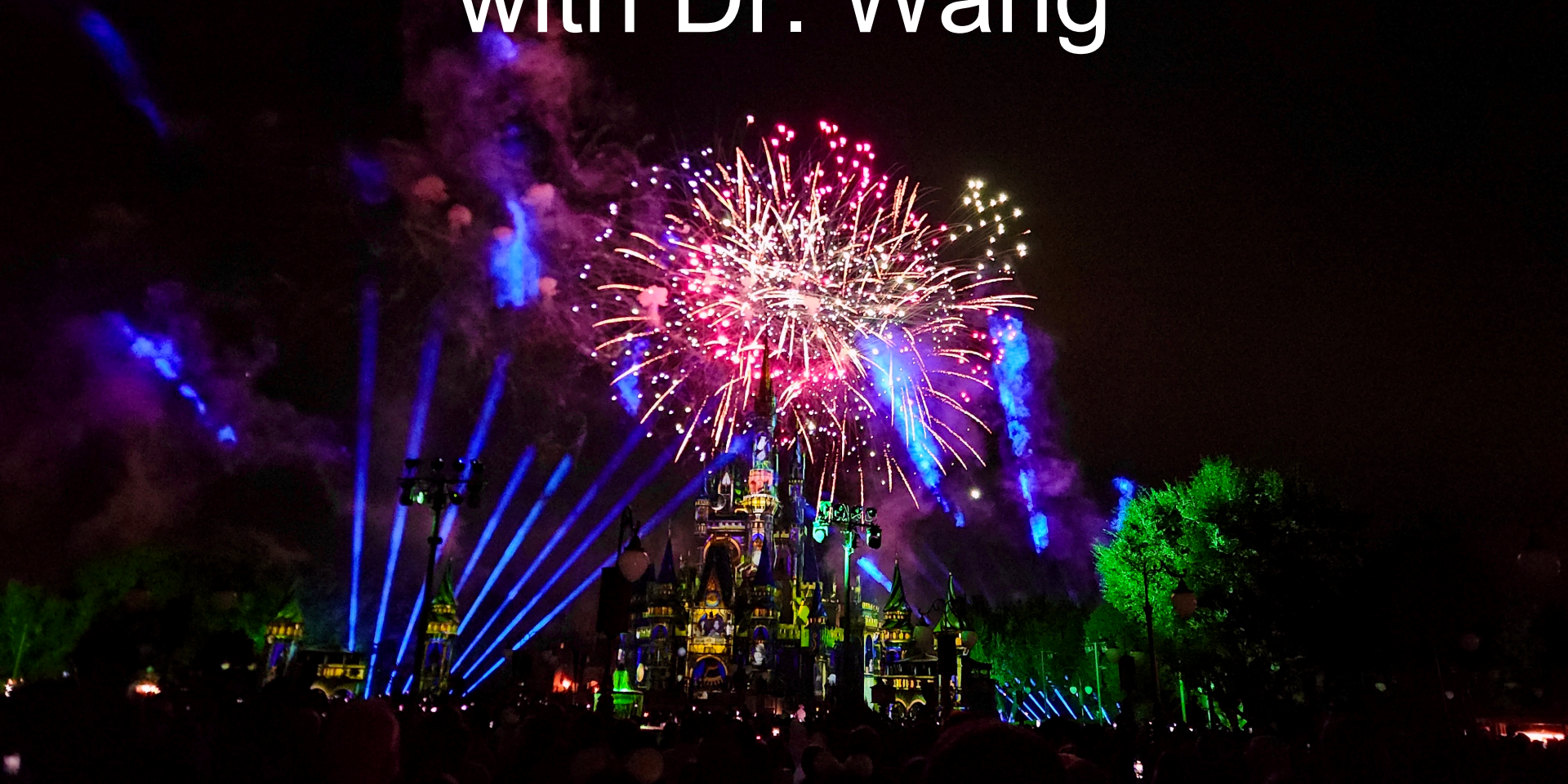


Welcome to Chem 150 with Dr. Wang



Please get a copy of Syllabus.

Introduction

Name: **Yiran (Ella) Wang / 王怿冉**

How to say: **yèe-ZHǎn wAAng**

Hometown: **Taiyuan, Shanxi, China**

You can call me: **Dr. Wang**

Education:

Undergraduate at Nankai University

B.S. in Chemistry (2017)



Introduction

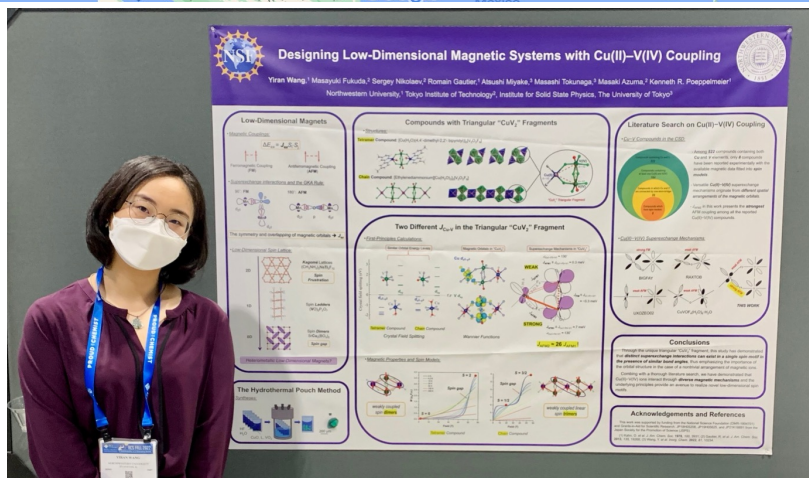
Education:

**Graduate at Northwestern University
Ph.D. in Chemistry (2022)**



New adventure:

**Instructional Assistant Professor
of Inorganic Chemistry (2023)**



Teaching is fun!

We are here to support you!

You can find Dr. Wang by...

- Dropping by Dr. Wang's office at Keck 231 to say hello!
- Emailing at yirawang@chapman.edu
 - Responding Hour: 9 AM-5 PM on weekdays
 - Typically respond within 24-h on weekdays
 - Weekend emails will be responded on Monday



Keck 231

We are here to support you!

You can find Dr. Wang by...

- **Office Hour**
 - Tue 10:00–11:00 AM at KC 230
 - Thu 10:00–11:00 AM (<https://chapman.zoom.us/j/93223626323>)
 - Fri 1:30–3:00 PM at KC 382
- **CHEM RePS (Review & Problem Solving) Sessions**
 - Wed 2:00 PM-3:30 PM at KC 230
- **Scheduling a 1-on-1 (or small group) meeting:**
<https://calendly.com/yiranellawang/1-on-1>



Keck 230 Conference Room



Keck 382

Paige Kwan

Hometown: Los Angeles, CA

Hobbies: crocheting, hiking, cafe and restaurant hopping, traveling, photography,

Year: Senior (C/O 2026)

Major: Biology

Minors: Chemistry and Sociology

Involvements: ASBMB, research, SI

Email: pakwan@chapman.edu



Tips for Succeeding in GenChem!

- Do practice problems consistently
- Form study groups
- Take advantage of additional resources
 - Professor and SI office hours
 - SI sessions
 - Tutoring in TLC
- Keep track of tentative dates for quizzes and exams
- Stay on top of assignments
- Complete additional problems sets



Link for potential SI session and
office hour times

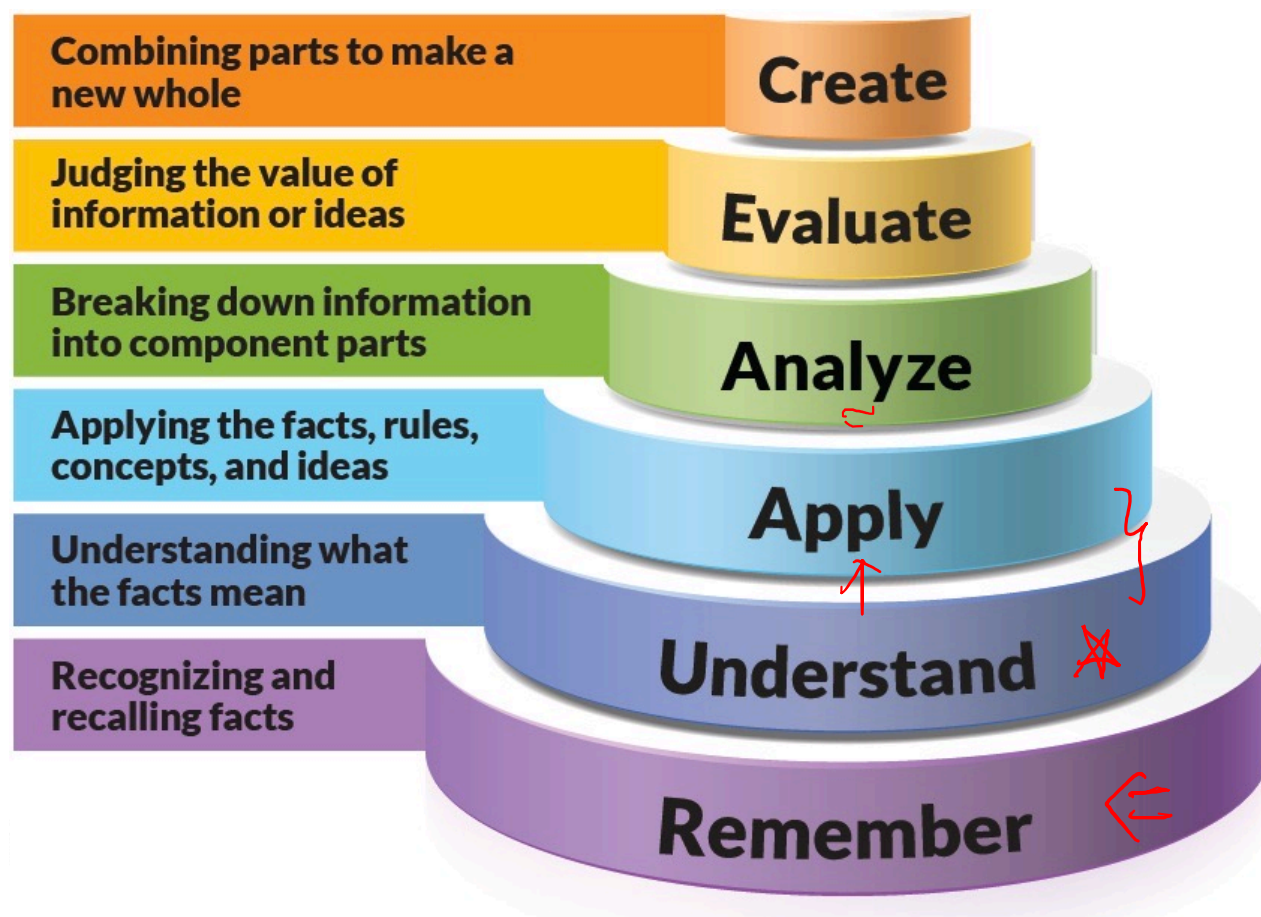
Ice Breaker (5 minutes)

A little bit of sharing

- Get into group of 2 or 3
- Introduce to each other with names you'd like to use in this class
- Share anything you want! (your hometown, your experience, your hobbies, fun thing you did, things you want to do ...)



Bloom's Taxonomy



- Constant work (Pre-Class Assignments; SI Session; ChemRePS Session)
- Frequent reflect on learning (Exam Wrapper Assignment)
- Study in groups
- Communicate with Dr. Wang OFTEN and EARLY

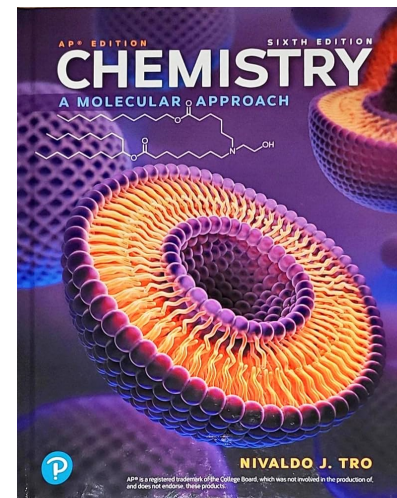
Chem 150 Canvas

- The **Syllabus** is your course contract
- Read **Canvas Announcements** after every class
 - Notification Settings
- Canvas Modules will provide your course resources
 - **Learning Objectives and Exam/Quiz Keys**
 - **Unit 1 Resources:**
 - Blank and Annotated Slides
 - Chem RePS Worksheets
 - Learning Objectives for Quizzes and Exams
 - **Unit 1 Assignments:**
 - Syllabus Quiz
 - Pre-Class Assignment Week 1-W

Course Materials

- Textbook (Tro: A Molecular Approach)

- e-textbook
 - 2 year access
- Hardback/Paperback/loose-leaf

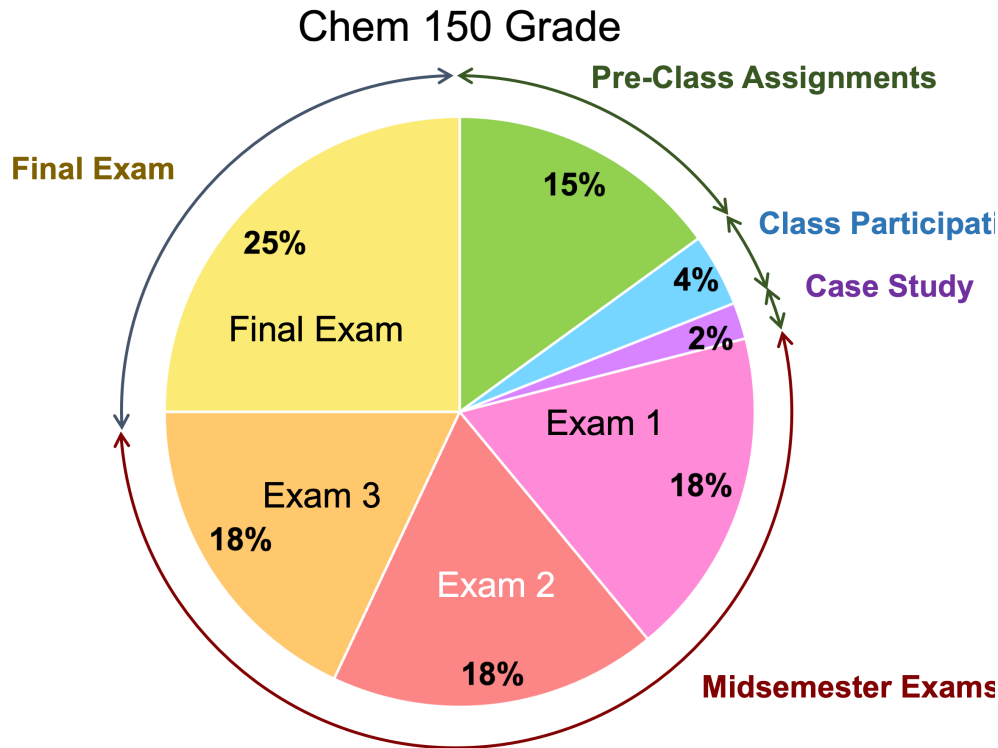


- OpenStax Chemistry: Atoms First 2e (free, online)

- **REQUIRED** online learning platform: Access Pearson

- Don't need to repurchase it as long as your access code didn't expire.
- If no access code or access code is expired: 14 days Temporary access
- **The 1st Assignment due Wednesday 2/4**

Evaluation and Assessment



Pre-Class Assignments (15%)

- Unlimited attempts
- The lowest 3 scores will be dropped.
- Due by **10 AM** on the day of the class
- Late assignments will incur a **10% deduction** for each day past the due date.
- Your grade will be synced after the due time.

Exams (79% in total)

- Three 50-minute midsemester exams (18% each) and a Final Exam (25%).
- Up to 4 bonus points can be added to each exam.

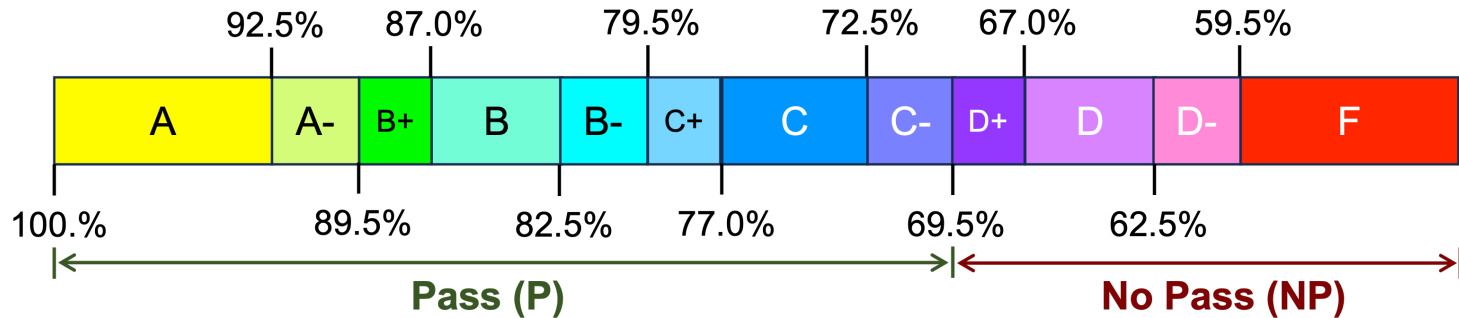
Class Participation (4%)

- Poll Everywhere
- logged in using your **Chapman email account**

Case-study (2%)

- Graded assignment in week 14

Evaluation and Assessment



Grade Scale

Grade	Points	Percentage		Grade	Points	Percentage
A	925-1000	92.5-100		C	725-769	72.5-76.9
A-	895-924	89.5-92.4		C-	695-724	69.5-72.4
B+	870-894	87-89.4		D+	670-694	67.0-69.4
B	825-869	82.5-86.9		D	625-669	62.5-66.9
B-	795-824	79.5-82.4		D-	595-624	59.5-62.4
C+	770-794	77.0-79.4		F	Below 595	Below 59.5

Resources and Policies

Academic Integrity

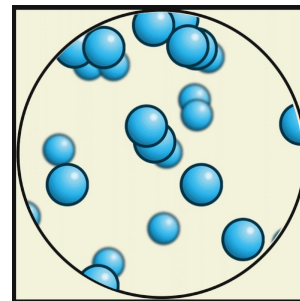
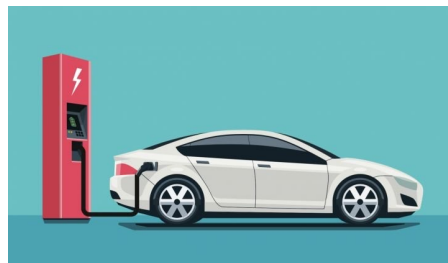
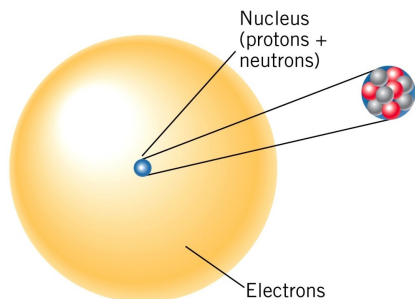
Student Support

Student with
Disabilities

Equity and
Diversity

Religious
Accommodation

Chem 140



Unit 1

Dimensional analysis,
Atoms & Element,
Compounds

Unit 3

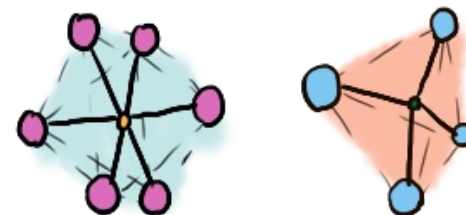
Redox Reaction,
Galvanic Cell,
Gas Law

Unit 2

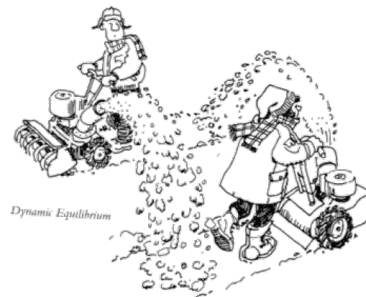
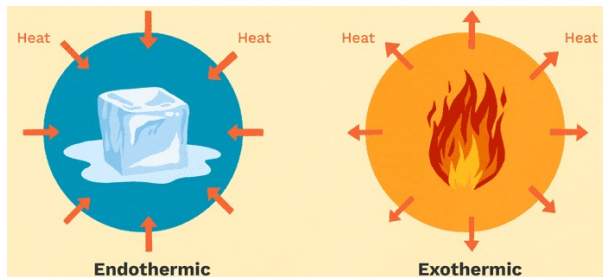
Chemical equation,
Aqueous Reaction

Unit 4 & 5

Quantum Mechanics,
Periodic Properties,
Chemical Bonding



Chem 150

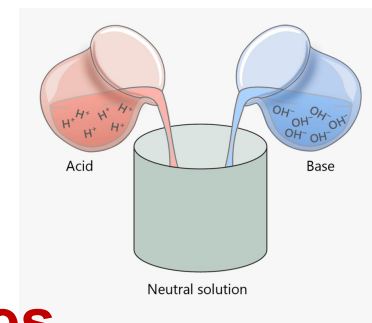
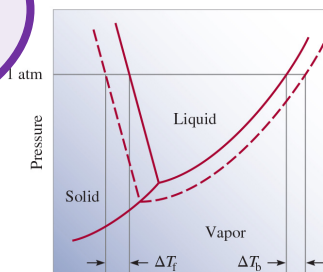
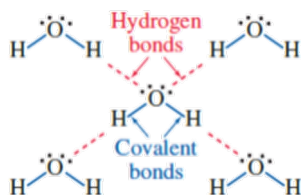


Unit 1
Energy, enthalpy,
Gibbs Free Energy

Unit 3
Kinetics, Chemical
equilibrium

Unit 2
Intermolecular Forces,
Phase of Matter

Unit 4
Acid & Base; Buffer



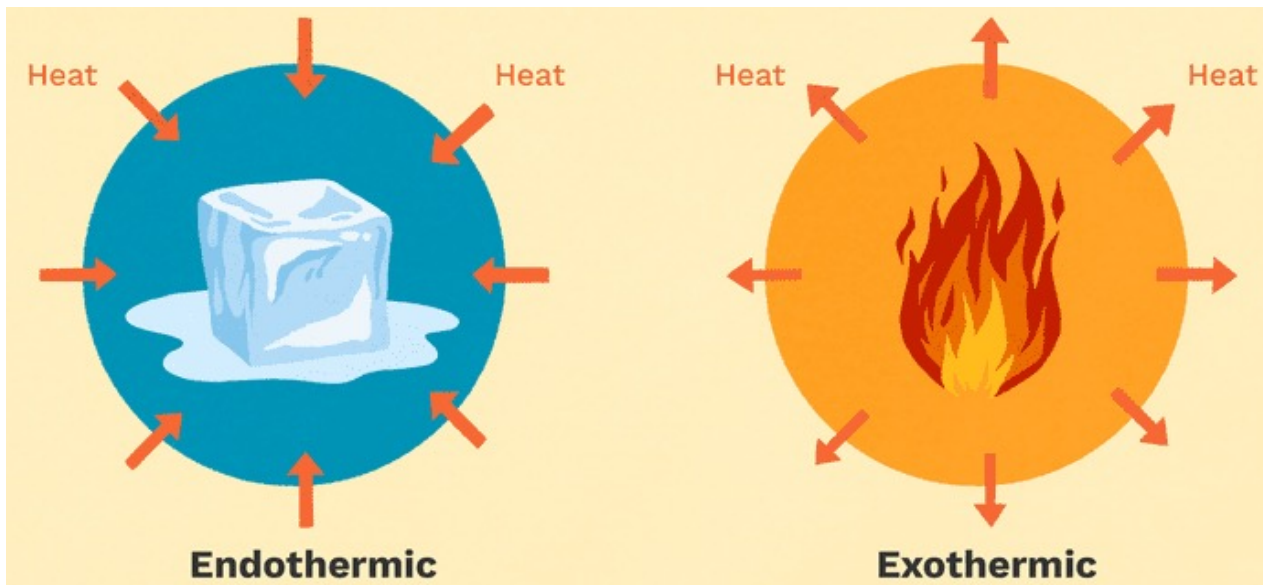
Structure
(How the particles come
together at micro-level)

Determine

Design

Properties
(How Matter behave at
macro-level)

Chapter 7: Thermochemistry



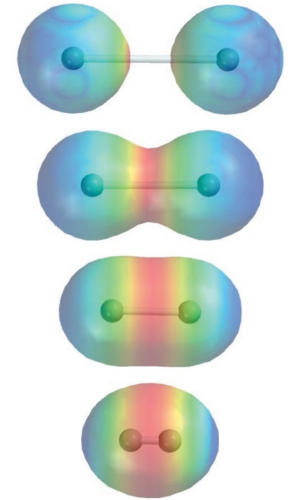
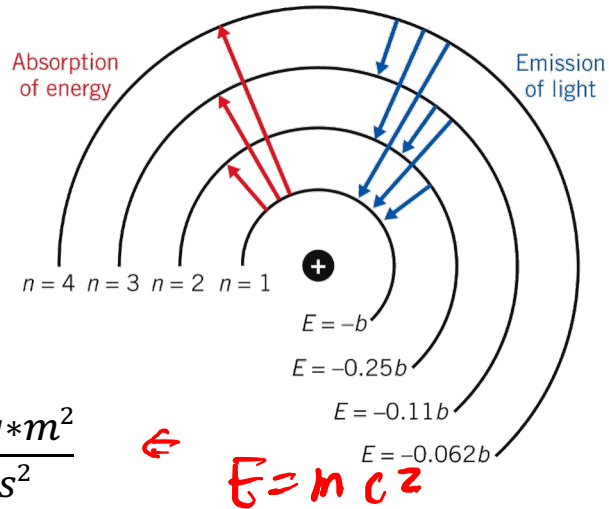
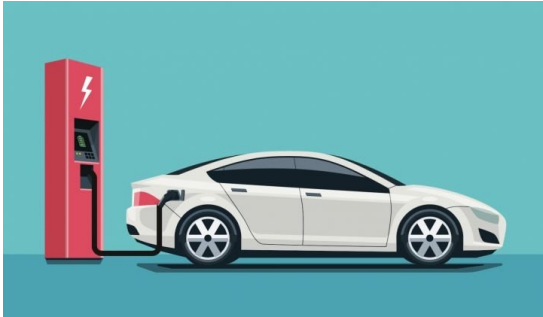
DR. WANG



Learning Objectives

1. Define energy and understand the nature of energy through the concepts of **heat** and **work**
2. Apply **Law of Conservation of Energy** by defining system and surroundings and tracking energy flow.
3. Identify energy changes as **exothermic** and **endothermic**
4. Understand the concept of **Internal Energy**
5. Calculate the change of **state function** based on conditions and chemical reactions

Energy (E)



Unit:

- Joule (J): $1 \text{ J} = 1 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2}$ $\Leftarrow$
- Calorie (cal): $1 \text{ cal} = 4.184 \text{ J (exact)}$ $\Leftarrow$

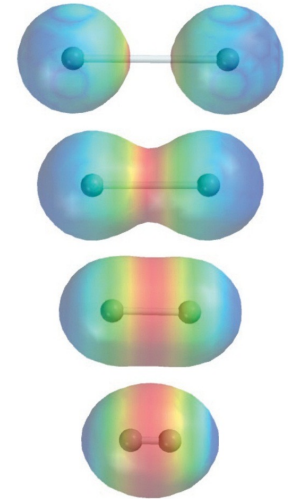
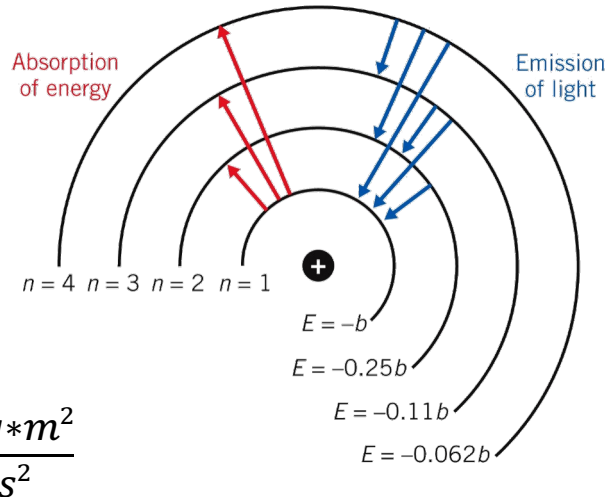
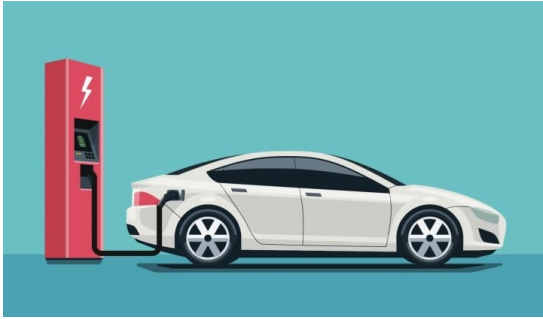
Definition:

- The capacity that an object possesses to do work
- Addable: Intensive / Extensive property

Classification:

- Kinetic Energy (KE)
- Potential Energy (PE)

Energy (E)



Unit:

- Joule (J): $1 \text{ J} = 1 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2}$
- Calorie (cal): $1 \text{ cal} = 4.184 \text{ J}$ (exact)

Definition:

- The capacity that an object possesses to do work
- Addable: Intensive / Extensive property $E = E_1 + E_2 + E_3 + \dots$

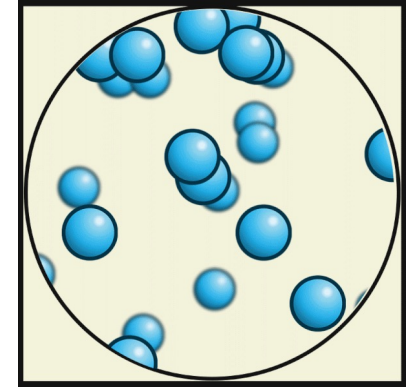
Classification:

- Kinetic Energy (KE)
- Potential Energy (PE)

Classification of Energy

Kinetic energy (KE)

- Associated with the motions of an object
- $KE = \frac{1}{2}mv^2$ (proportional to mass and Velocity)
- At molecular level:



Average Kinetic energy of Molecular Motions = thermal energy

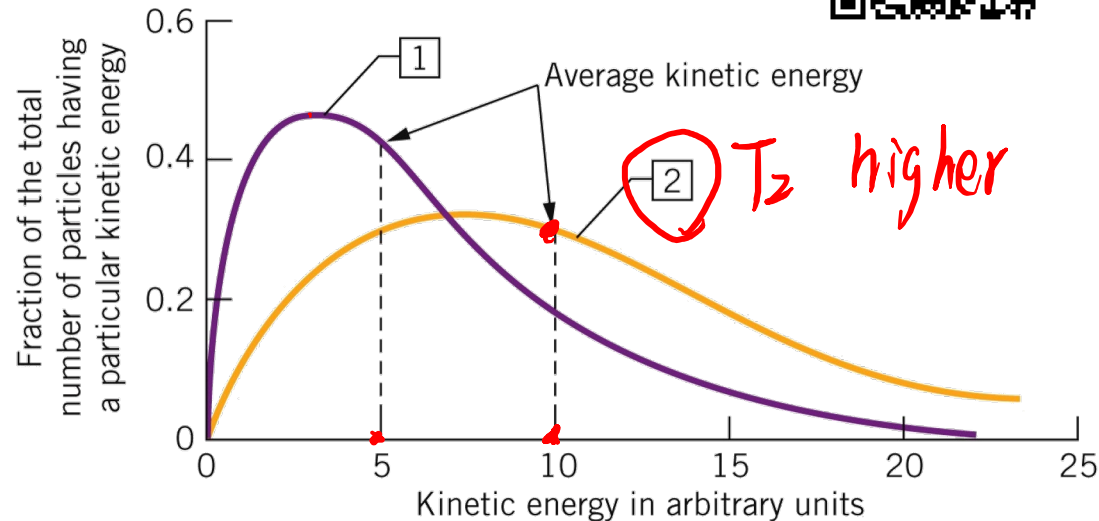
- Absolute zero (the coldest temperature possible):
At 0 K (Kelvin), molecular motion virtually stops.



- Higher temperature means:

higher thermal Energy;

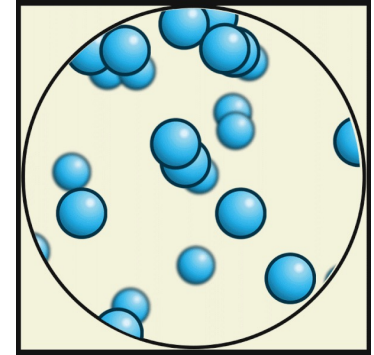
faster molecular motion



Classification of Energy

Kinetic energy (KE)

- Associated with the motion of an object
- $KE = \frac{1}{2}mv^2$ (proportional to mass and speed)
- At molecular level:



Average Kinetic energy of Molecular Motions = thermal energy

- Absolute zero (the coldest temperature possible):

At 0 K (Kelvin), molecular motion virtually stops.

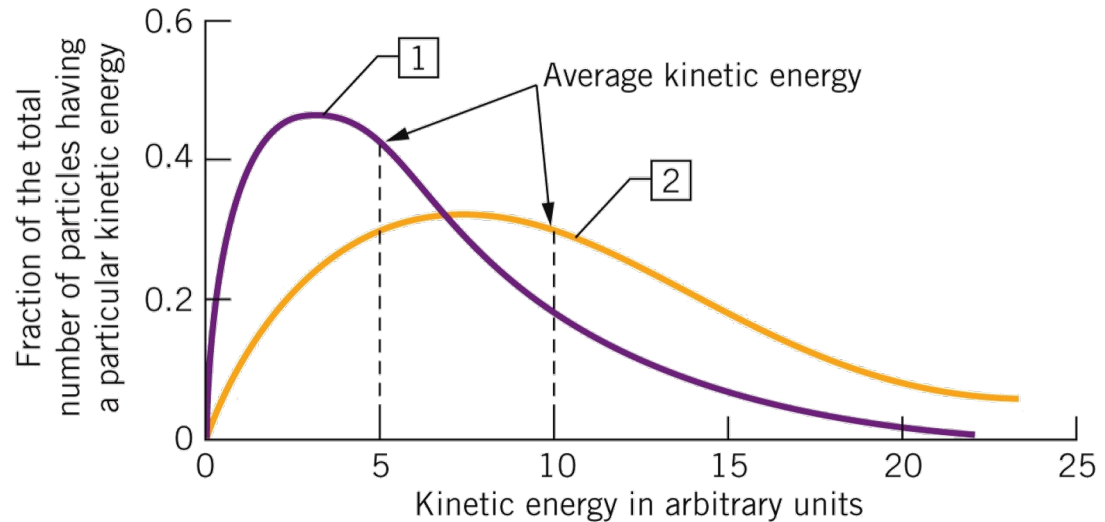
- Higher temperature means:

higher thermal Energy;

faster molecular motion

$$KE_1 < KE_2$$
$$T_1 < T_2$$

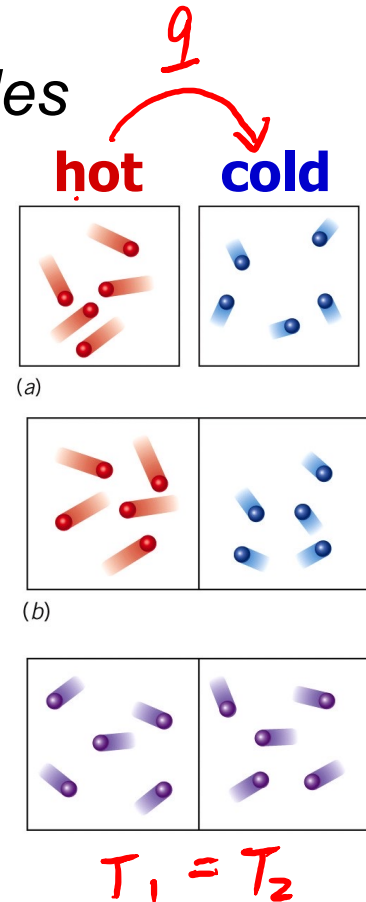
(poll it)



Classification of Energy

Thermal Energy: Average Kinetic energy of Particles

- **Temperature (T)** measures the average kinetic energy of atoms/ molecules
- Temperature is Intensive / Extensive property
- **Heat (q)**: the transfer of thermal energy between two bodies that are in contact and are at different temperatures
= Energy
- Heat is Intensive / Extensive property
- **Thermal Equilibrium:**



Overtime, heat transfers from the object with higher temperature to the object with lower temperature until the two objects have the same temperature (same average kinetic energies).

At thermal equilibrium, $q_{\text{release}} + q_{\text{absorb}} = 0$ and $T_1 = T_2$. $q_{\text{release}} = -q_{\text{absorb}}$ (hot) (cold)

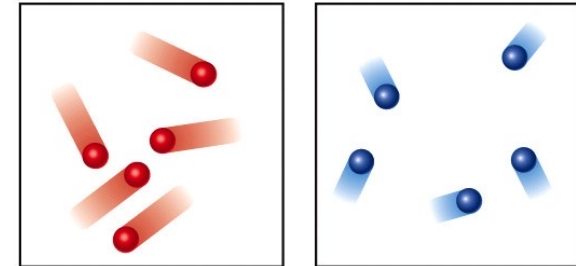
Classification of Energy

Thermal Energy: Average Kinetic energy of Particles q

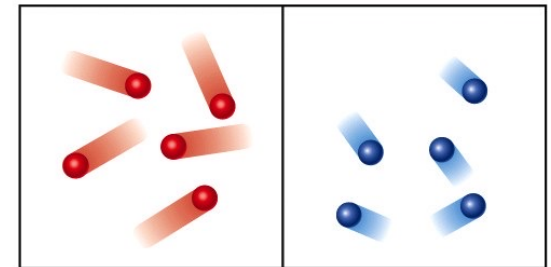
- **Temperature (T)** measures the average kinetic energy of atoms/ molecules
- Temperature is Intensive / Extensive property
 $T \neq T_1 + T_2$
- **Heat (q)**: the transfer of thermal energy between two bodies that are in contact and are at different temperatures
- Heat is Intensive / Extensive property
 $q = q_1 + q_2$
- **Thermal Equilibrium:**

Overtime, heat transfers from the object with higher temperature to the object with lower temperature until the two objects have the same temperature (same average kinetic energies).

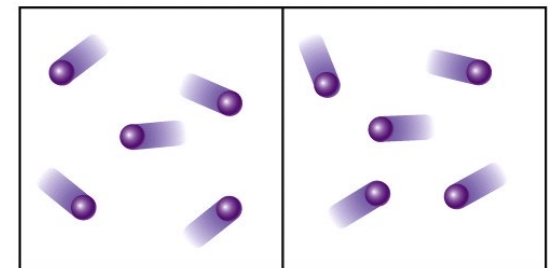
hot $\Rightarrow$ cold



(a)



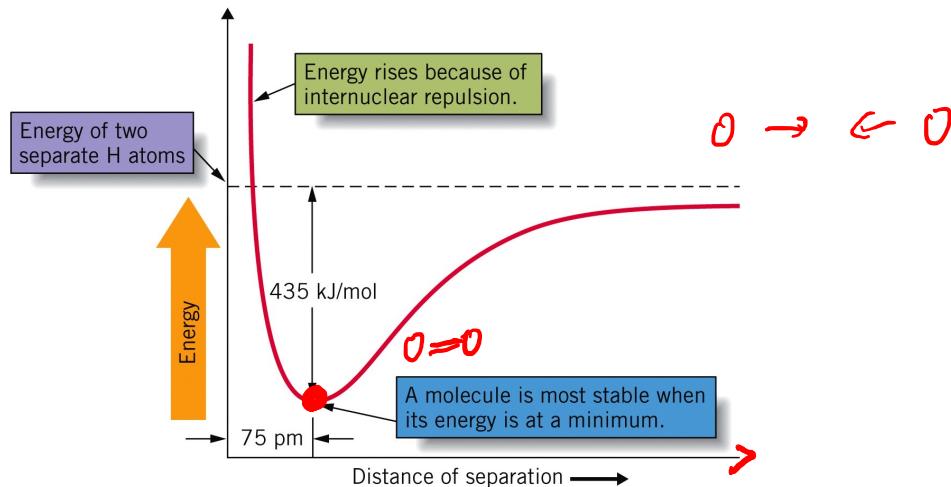
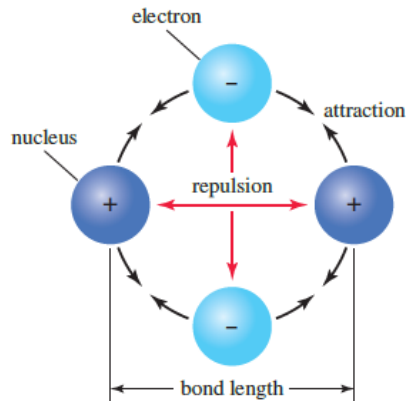
(b)



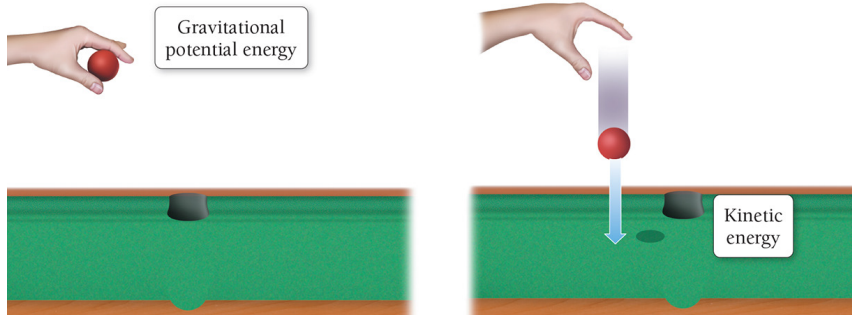
Classification of Energy

Potential energy (PE)

- Associated with the position or composition of an object (chemical energy)
- Gravitational, Electric, Mechanical (Spring)...
- Chemical Potential Energy: Stored in chemical bonds



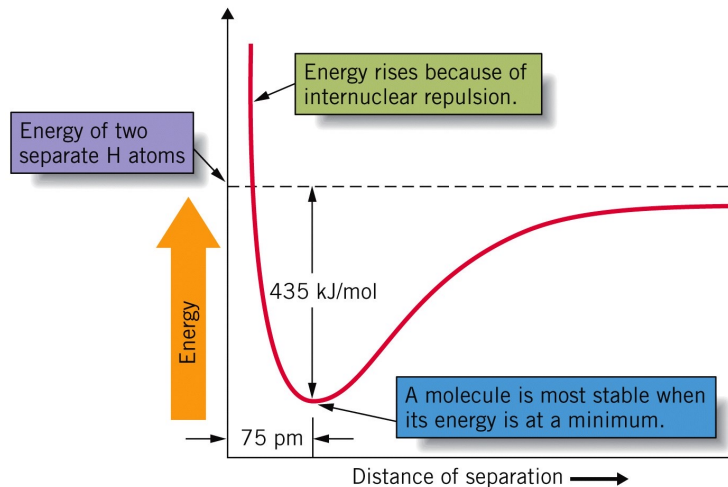
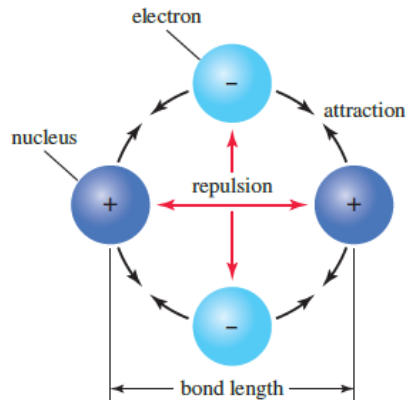
- Nature prefers lowest potential energy: most stable



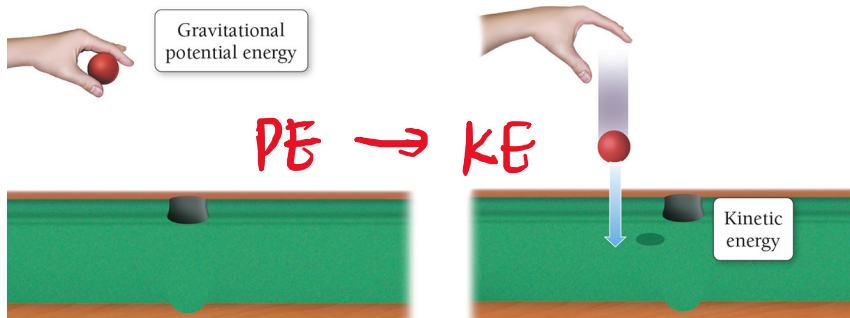
Classification of Energy

Potential energy (PE)

- Associated with the position or composition of an object
- Gravitational, Electric, Mechanical (Spring)...
Altitude, charge $\oplus \ominus$;
- Chemical Potential Energy: Stored in chemical bonds



- Nature prefers lower potential energy: most stable



*PE $\uparrow$:
Altitude $\uparrow$
repulsion $\uparrow$, attraction $\downarrow$
Pressed spring.*

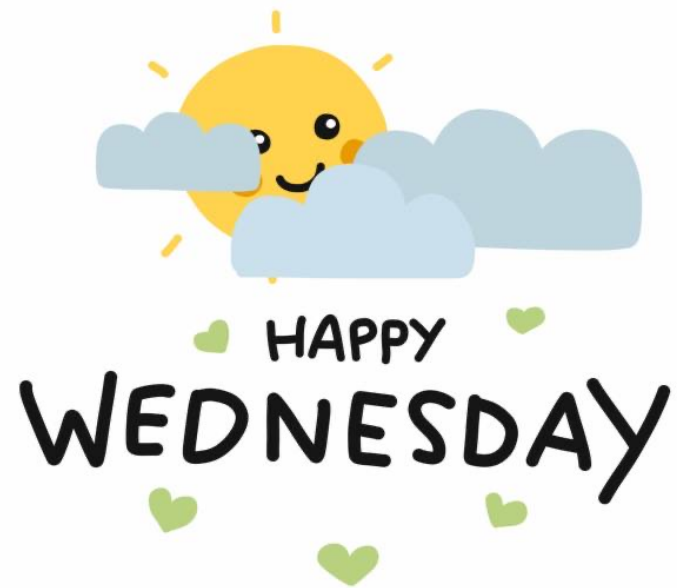
Announcement 2/4

Assignments that are due soon

- Pre-Class Assignment Week 1-W
- Pre-Class Assignment Week 1-F

Office Hour

- Dr. Wang's: Thu 10-11 AM on Zoom



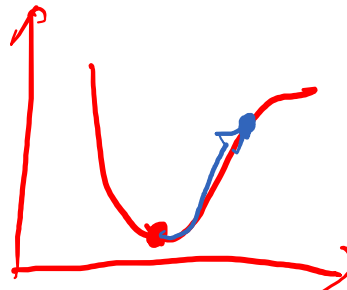
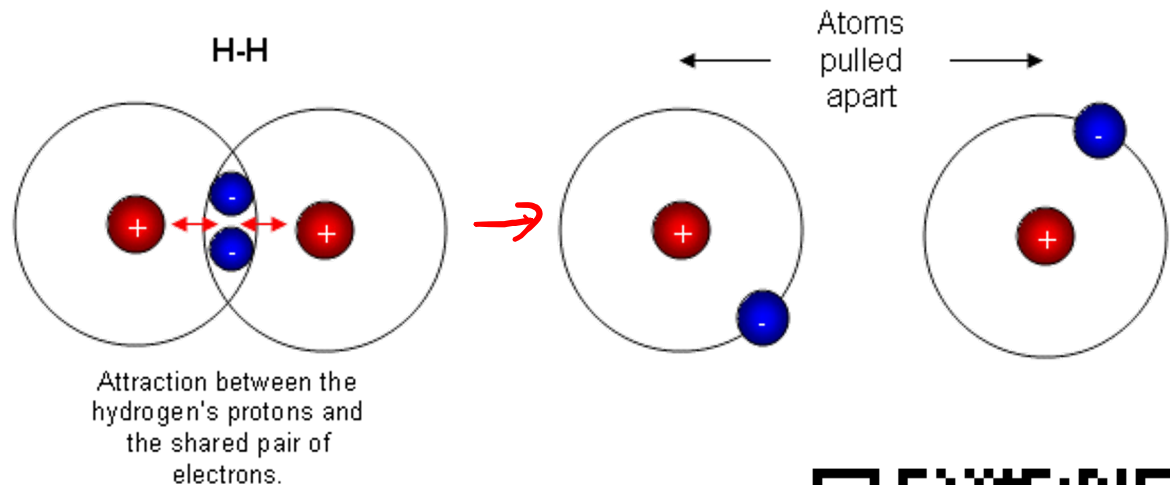
Your Turn!

In the following picture depicts the process of a chemical bond breaking between the two H atoms. During this process, the chemical potential energy between the H atoms:

A. Increases

B. Decreases

C. Stay the same



System & Surroundings

- **System:** the body of your interest
- **Surroundings:** Everything else
- Three types of systems:

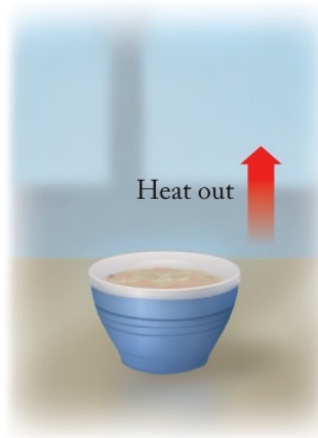
Energy exchange? X
Matter exchange? X



isolated system

(only exists in theory)

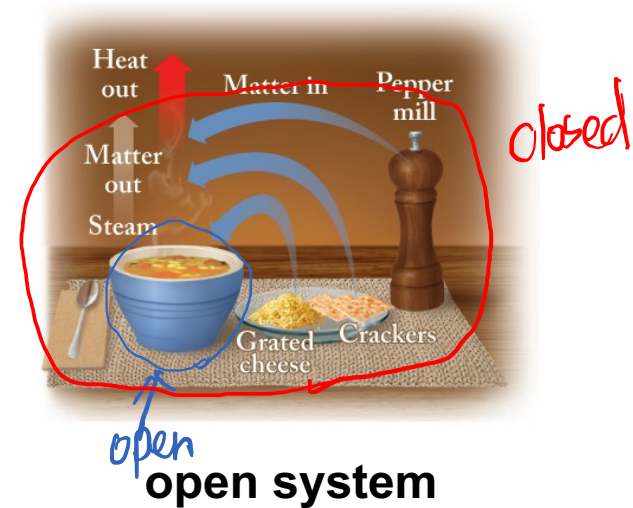
Energy exchange? ✓
Matter exchange? X



closed system

(most common)

Energy exchange? ✓
Matter exchange? ✓



- The energy analysis will change for the same scenarios if the system is changed.

Exothermic vs Endothermic Processes

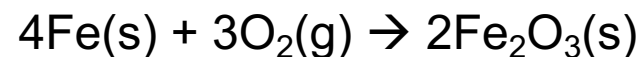
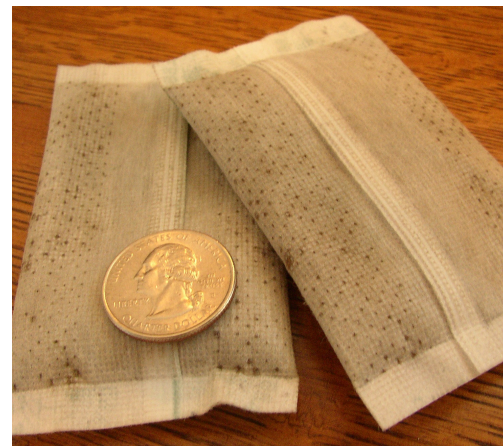
With the perspective of the “system”

- **Exothermic** processes lose heat (-): $q < 0$
- **Endothermic** processes gain heat (+): $q \geq 0$

Ex. A hot pack is used to warm up your cold hands. Is this process exothermic or endothermic? What is the flow of energy? (the “system” is underlined)

	System	Surroundings
Object	hot pack	hands
Final (temp)	warm (30°C)	warm (30°C)
Initial (temp)	hot (100°C)	cold (20°C)
Change (heat) = Final - Initial	(-) lose	(+) gain
Exo/Endo- thermic	exothermic	endothermic

q



Your Turn!

Which of the following is an **endothermic** process from the perspective of the system? In each case, the “system” is underlined.

A. A bottle of wine being cooled in the cellar

B. A candle burning at a dinner table

C. A can of beer being chilled in a cooler



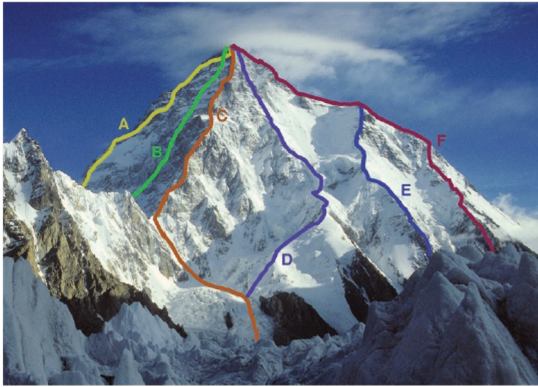
Exothermic: $q < 0$

Endothermic: $q > 0$

	A		B		C	
	System	Surr.	System	Surr.	System	Surr.
Object	wine		Candle	air	cooler	beer
Final (temp)	cool			hot		cool (10°C)
Initial (temp)	warm			warm		warm (30°C)
Change (heat) = Final - Initial	(-)			(+)		(-)
Exo/Endo- thermic	exo-		exo-	endo-	endo-	exo-

Thermodynamics and State Function

- **Thermodynamics**: study of energy and its interconversions
- Energy is a “**state function**” (the change only depends on the initial and final conditions, NOT on the process used).



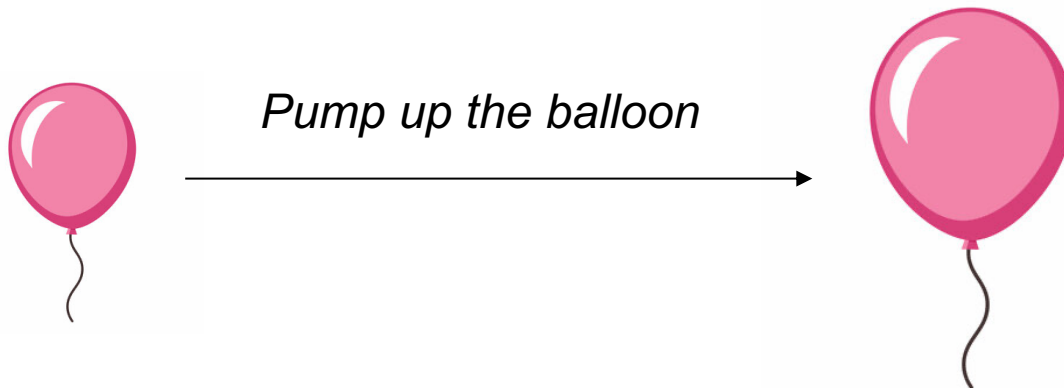
$$\Delta E = E_{\text{final}} - E_{\text{initial}}$$

- State function is to determine whether the path taken to reach the function affects the value. **X path doesn't matter**
(Imagine taking different pathways to climb the mountain.)
- Is work a state function? **X not a state function**
- Is temperature a state function? **✓ is a state function**

Your Turn!

Is volume a state function?

- A. Yes it is a state function.
- B. No it is not a state function.
- C. It depends on the situation.



State function is to determine whether the path taken to reach the function affects the value.

The First Law of Thermodynamics

- Internal Energy of a System is the sum of the chemical kinetic and potential energies of all particles that compose the system.

$$E_{\text{Internal}} = \sum \text{PE (potential energy)} + \sum \text{KE (kinetic energy)}$$

- The absolute internal energy of a given system cannot be known but the change (ΔE) can be tracked: $\Delta E = E_{\text{final}} - E_{\text{initial}}$

$\Delta E < 0$, the sign of ΔE is - and the system lose energy

$\Delta E > 0$, the sign of ΔE is + and the system gain energy

The First Law of Thermodynamics

The law of conservation of energy:

- The total energy of the universe is **constant**.
- Energy, like matter, cannot be destroyed or created

$$E_{\text{final_system}} + E_{\text{final_surroundings}} = E_{\text{initial_system}} + E_{\text{initial_surroundings}}$$


$$E_{\text{final_system}} - E_{\text{initial_system}} + E_{\text{final_surroundings}} - E_{\text{initial_surroundings}} = 0$$

$$\Delta E_{\text{system}} + \Delta E_{\text{surr.}} = 0$$

- If $\Delta E_{\text{system}} > 0$ (gain energy), then $\Delta E_{\text{surroundings}} < 0$ (lose energy)
- If $\Delta E_{\text{system}} < 0$ (lose energy), then $\Delta E_{\text{surroundings}} > 0$ (gain energy)

Your Turn! $\Delta E = E_{\text{final}} - E_{\text{initial}}$

A system initially has a total energy of 225 J. After an energy exchange with the surroundings, the system has 256 J of energy. What was the energy change of the surroundings? Did the surroundings absorb or release the energy?

- A. + 31 J; Absorb
- ~~B. - 31 J; Absorb~~
- ~~C. + 31 J; Release~~
- D. - 31 J; Release



	System	Surroundings
E(Final)	256 J	
E(Initial)	225 J	
ΔE	+31 J	-31 J
Absorb/Release Energy	absorb	release

= 0